

Introduction to Statistical Methods - Mid-Semester 1 Mock 01

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ISM Midsem 1 Mock 01 - Question Paper

Course: AMLCS ZC418 Introduction to Statistical Methods

Time: 90 minutes

Marks: 30

Paper difficulty: Standard Scope: Sessions 1-8

For every calculation-bearing numerical or applied response, show the governing formula, substitution of supplied values, non-obvious intermediate steps, units or domain and validity conditions, and interpretation. For a qualitative applied response, give a reasoned not applicable for formula, substitution, and calculation where appropriate; assumptions or conditions and interpretation remain required.

Q1 - Population descriptions - 2 marks

Sessions: 1 | Topics: population, sample, parameter, statistic

Point difficulty mix: Easy 2, Standard 0, Stretch 0 Provenance: Authored BITS-style | Time guidance: 5 minutes

A city authority records journey times from 120 randomly selected bus journeys to learn about all bus journeys operated that month. Define parameter and statistic, then identify one parameter and its corresponding statistic in this context.

Q2 - Two probability distinctions - 2 marks

Sessions: 2 | Topics: exclusive versus independent; permutation versus combination

Point difficulty mix: Easy 1, Standard 1, Stretch 0 Provenance: Authored BITS-style | Time guidance: 6 minutes

Compare mutually exclusive events with independent events, including their mathematical tests. Then compare a permutation with a combination and give one condition that determines which count applies.

Q3 - Table denominators - 4 marks

Sessions: 3 | Topics: joint, marginal, conditional probability, independence

Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 12 minutes

The observed shipment counts are:

Route	Total	Delayed (D)
R	80	32
S	120	18

There are 200 shipments in total.

Calculate $P(R \cap D)$, $P(D)$, and $P(R|D)$, and determine whether R and D are independent. Include the formula, substitutions, intermediate totals, probability domain and denominator conditions, and contextual interpretations.

Q4 - Naive Bayes classification - 4 marks

Sessions: 4 | Topics: prior, conditional feature likelihood, normalization, model caveat

Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 13 minutes

A classifier has classes Urgent and Routine with priors 0.30 and 0.70. For a message having features short=1 and keyword=1, the class-conditional probabilities are:

Class	Short-feature likelihood	Keyword-feature likelihood
Urgent	0.80	0.60
Routine	0.25	0.20

State the within-class conditional-independence meaning. Calculate both scores, their normalizer, both posteriors, and the selected class; state probability conditions and one caveat.

Q5 - PMF normalization and moments - 4 marks

Sessions: 5 | Topics: PMF, expectation, variance

Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 13 minutes

Let X be the number of support incidents in one hour. For $x=0,1,2$, it has PMF $p(x)=k(x+1)$. Find k , $E[X]$, $E[X^2]$, and $\text{Var}(X)$. Show substitutions/fraction steps; verify PMF validity; state support/units; interpret both moments.

Q6 - Binomial reliability count - 4 marks

Sessions: 6 | Topics: binomial probability and moments

Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 12 minutes

Eight independent sensors each fail with probability 0.25; X counts failures. Choose and justify the distribution using support/all trial conditions. Calculate $P(X=2)$, $E[X]$, and $\text{Var}(X)$ with formula, substitution, and arithmetic; interpret each result. Keep checks distinct: support/trial conditions justify the model; bounds, nonnegative variance, units, and repeated-group meaning check/interpret results.

Q7 - Mean confidence interval decision - 3 marks

Sessions: 7 | Topics: unknown-sigma t interval, conditions, frequentist interpretation

Point difficulty mix: Easy 0, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 11 minutes

An independent random sample of 36 processing times has mean 72.0 seconds and sample standard deviation 9.0 seconds; the population standard deviation is unknown. A normal-population model is considered reasonable. Use the supplied 95% critical value $t^*=2.030$ with 35 degrees of freedom. Choose and construct the appropriate two-sided interval for the population mean. Show formula, substitutions, standard error, margin, units, design/model conditions, and a correct frequentist interpretation.

Q8 - Integrated method audit - 2 marks

Sessions: 8 | Topics: method selection, conditions, interpretation

Point difficulty mix: Easy 0, Standard 1, Stretch 1 Provenance: Authored BITS-style | Time guidance: 6 minutes

A manager says, "We observed an average of 5 network incidents per day, so the number in the next three days is binomial with $n=3$." Evaluate this model choice. Give a reasoned not applicable for calculation steps, identify the missing assumptions and a more plausible model if stated conditions held, and explain how exposure and support would be handled.

Q9 - Quartile and checking properties - 3 marks

Sessions: 1, 8 | Topics: quartile convention, fences, whiskers, skewness checks

Point difficulty mix: Easy 1, Standard 2, Stretch 0 Provenance: Authored BITS-style | Time guidance: 6 minutes

State the median-of-halves quartile convention used in this package. Define IQR and the two 1.5 IQR fences. Then distinguish calculated fences from observed box-plot whiskers and state two cautions about interpreting Pearson's second skew coefficient.

Q10 - Replacement and event relations - 2 marks

Sessions: 2 | Topics: replacement, independence, mutually exclusive events

Point difficulty mix: Easy 1, Standard 1, Stretch 0 Provenance: Authored BITS-style | Time guidance: 6 minutes

Compare two draws with replacement against two draws without replacement from a finite urn. Explain how later probabilities change, then distinguish that dependence question from whether two events on the same draw are mutually exclusive. Give one valid same-draw example of two mutually exclusive events.

End of paper - 30 marks